

## Aircraft Design Requirements (2026-2027)

### **Performance Requirements:**

- Max takeoff distance: 100ft
- Max landing distance: 400ft
- Flight attempt time limit: 60s
- Aircraft must perform a full flight circuit/lap within the attempt limit. Takeoff, fly 400ft before the first 180 degree turn, fly back, turn another 180 degrees, then land.

### **Design Requirements:**

- Maximum MTOW: 55 pounds
- Wing span must be greater than 6 feet but less than 8 feet. Minimum wing chord is greater than 4 inches
- Body/fuselage length must be less than 12 feet
- No use of fiber-reinforced plastic
  - o Exceptions: Off the shelf FRP Motor Mounts, propeller, landing gear, control linkages
- No lead anywhere

### **Avionics/Propulsion Requirements:**

- Circuit MUST include red arming plug, at least 9in away from any propeller and easily accessible to cut power
- Separate battery is required for receiver system and must operate even without the arming plug installed
- Must use 2.4 GHz radio system
- 2 or 4 motors
- Powered by a commercially available 4 cell (14.8V) Li-Po battery pack, max of 2200 mAh
- No metal propellers

### **Payload:**

- Commercially available 2 liter bottles as payload. Empty bottle is at least 1 pound, filled bottle is at least 4 pounds
- Must increase payload after each successful flight attempt
- Cargo cannot be exposed to air or shift in flight
- Payload must be unloaded within 1 minute by two team members

## Scoring:

### 7.6 REGULAR CLASS SCORING

The team's Final Flight Score (FFS) is the average of the team's top three (3) flight scores achieved during the competition ( $FS_1$ ,  $FS_2$ , and  $FS_3$ ).

#### Scoring Equation:

$$FFS = \text{Final Flight Score} = (FS_1 + FS_2 + FS_3)/3 + PPB$$

#### Where:

$$FS = \text{Flight Score} = 3(EB) + 11(FB)$$

$$PPB = \text{Payload Prediction Bonus} = \text{MAX}(10 - (FS - PS)^2, 0)$$

$EB = \text{Count of EMPTY BOTTLES (\#)}$

$FB = \text{Count of FILLED BOTTLES (\#)}$

$PS = \text{Predicted Maximum Flight Score (\#)}$

The Predicted Maximum Flight Score (PS) is determined from the flight score prediction curve provided in the TDS (Section 4.5) and the density altitude measured at the event.

The Payload Prediction Bonus (PPB) will be calculated once at the end of the competition. The PPB will be calculated using whichever of the top three (3) flight scores utilized in the FFS calculation generates the largest PPB. A PPB score less than zero (0) will default to zero (0).

Ties in FFS will be broken by total number of filled bottles carried in top 3 scored flights.

#### Penalty Points

Penalty points assessed during competition are deducted from a team's overall score.

## Notes:

- For initial sizing, an ideal cruising speed needs to be calculated so we fall within the allotted timeframe.